

Enzyme Design

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AI.zymes: A Modular Platform for Evolutionary Enzyme Design

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Abstract: The ability to create new-to-nature enzymes would substantially advance bioengineering, medicine, and the chemical industry. Despite recent breakthroughs in protein design and structure prediction, designing novel biocatalysts remains challenging. Here, we present AI.zymes, a modular platform integrating cutting-edge protein engineering algorithms within an evolutionary framework (<https://github.com/bunzela/AIzymes>). By combining bioengineering tools such as Rosetta, ESMFold, ProteinMPNN, and FieldTools in iterative rounds of design and selection, AI.zymes can optimize a broad range of catalytically relevant properties. In addition to enhancing transition state affinity and protein stability, AI.zymes can also improve properties that are not targeted by the employed design algorithms. For instance, AI.zymes can enhance electrostatic catalysis by iteratively selecting variants with stronger catalytic electric fields. Benchmarking AI.zymes on the promiscuous Kemp eliminase activity of ketosteroid isomerase led to a 7.7-fold activity increase after experimentally testing just 7 variants. Due to its modularity, AI.zymes can readily incorporate emerging design algorithms, paving the way for a unifying framework for enzyme design.

Introduction

Enzymes are nature's catalysts, facilitating chemical reactions with remarkable specificity and efficiency. The ability to create tailor-made enzymes holds immense potential for biotechnology, medicine, and the chemical industry.^[1,2] However, the vastness of sequence space and the complex relationship

between structure and function make enzyme design and engineering a significant challenge. Directed evolution is a powerful method for navigating sequence space and optimizing enzyme activity in iterative rounds of mutagenesis and selection.^[3–5] Yet, laboratory evolution is time-consuming and struggles to access distant regions of the fitness landscape.

Computational protein design provides an alternative to directed evolution for generating functional proteins with tailored properties. In contrast to directed evolution, protein design typically relies on introducing mutations into a fixed set of input structures, limiting diversity and access to distant functional variants.^[6–10] Evolutionary design overcomes this limitation by iterating through cycles of design and selection, using the best-performing variants as templates for further optimization.^[11,12] By continuously varying the input structure, evolutionary approaches can access sequences that are inaccessible in a single design cycle (Figure 1a). Although computational evolution has been applied to protein and nucleic acid binders,^[11,12] its use in enzyme design has been limited by challenges in accurately modeling and scoring biocatalysts.

Recent protein design and structure prediction breakthroughs have substantially boosted our enzyme engineering abilities. Tools such as AlphaFold3 and ESMFold can be used to predict protein structures to inform engineering,^[13,14] and generative models such as ProteinMPNN and RFdiffusion can afford de novo proteins, especially when targeting simple and well-defined folds.^[8,9] These advances recently culminated in the design of several enzymes by creating novel protein scaffolds around active-site models.^[15,16]

Even the best designer enzymes fall short of the catalytic power of natural enzymes.^[15,16] Most protein design algorithms are developed primarily to enhance protein stability rather than to optimize biocatalytic properties.^[6–10] As a result, computational design often prioritizes active-site stability while overlooking electrostatic and dynamic factors essential for biocatalysis.^[3,17–19] Effects such as electrostatic catalysis^[20,21] and conformational dynamics^[3,17,18] are typically addressed only after design by filtering variants.^[22,23] This methodological gap likely contributes to the low k_{cat} values of de novo enzymes,^[15,16] underscoring the need for a holistic multiobjective framework that integrates biocatalytically relevant properties into the design process.

To address this gap, we developed AI.zymes, a modular platform for evolutionary design that integrates multiple bioengineering tools, such as Rosetta, ESMFold, ProteinMPNN, and FieldTools into a coherent design pipeline (Figure 1b). AI.zymes was benchmarked on the promiscuous Kemp eliminase activity of ketosteroid isomerase. Optimization of transition-state recognition, protein stability, and catalytic

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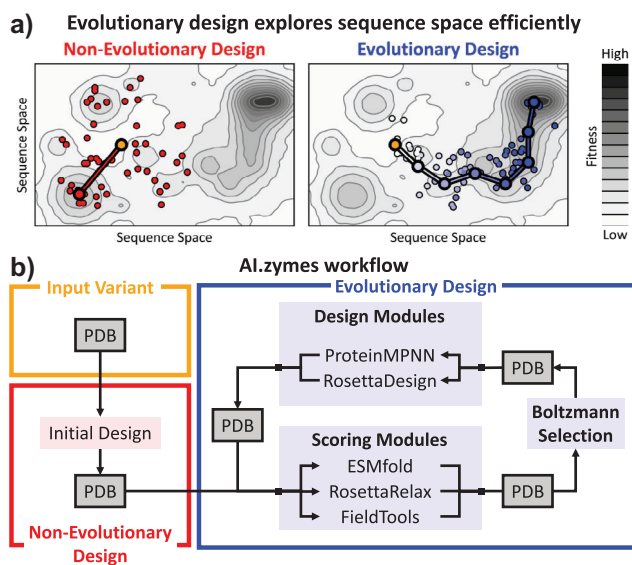


Figure 1. Evolutionary design with AI.zymes. a) Non-evolutionary design (red) starts from a fixed input variant (yellow), which limits diversity. In contrast, evolutionary design (blue) explores sequence space more efficiently by varying the input sequence. b) Modularity in AI.zymes is achieved by linking standalone algorithms through their output structures (PDB, grey).

electric fields resulted in a 7.7-fold activity improvement after testing only 7 variants.

Results and Discussion

How AI.zymes Works

Similar to non-evolutionary design, AI.zymes begins by generating an initial pool of variants from a fixed input structure. Subsequently, evolutionary design is performed by iteratively selecting variants from the growing design pool for redesign (Figures 1 and S1). Unlike “rounds”-based evolution, AI.zymes operates on a continuously expanding variant pool, selecting and redesigning variants regardless of their generation.

During selection, AI.zymes first eliminates variants that do not align with the target geometry between the chemical transition state and catalytic residues.^[24] Then, multiobjective Boltzmann selection is performed based on the ligand–protein interaction strength (interface score),^[25] protein stability (stability score), and electrostatic catalysis (electric fields).^[19] AI.zymes adopts a forward-looking evolutionary approach, selecting variants based on their “potential,” which is calculated as the mean of a variant’s score and the scores of its immediate descendants. To achieve multiobjective optimization, Boltzmann selection is based on the average of the standardized potentials. Finally, AI.zymes gradually adjusts the Boltzmann temperature during selection to transition from an exploration phase, where input structures are chosen more randomly, to an exploitation phase that is highly selective for the most promising variants.

Following selection, variants are subjected to structure prediction with ESMFold^[13] and relaxation using RosettaRelax^[7,26] to assess conformational stability and backbone shifts. Once relaxed, the resulting structures undergo redesign. In the current implementation of AI.zymes, active-site residues are optimized with RosettaDesign,^[8,9] whereas the protein scaffold is targeted using ProteinMPNN^[9] (Figure 2b). This division leverages the strengths of each algorithm: Rosetta readily accounts for chemical ligands, whereas ProteinMPNN excels at enhancing protein stability. To combine the design methods, each is applied with a user-defined probability during redesign (RosettaDesign: 70%; ProteinMPNN: 30%).

AI.zymes is built to be highly modular and operates on modules that use PDB files as input and output (Figure 1b). For ProteinMPNN, this is achieved by linking the designed sequence to structure prediction and relaxation. The modular structure of AI.zymes facilitates the integration of additional design methods and ensures the continued applicability of AI.zymes in the rapidly evolving field of computational protein design.

To explore sequence space gradually, AI.zymes constrains the mutational load per design step (Tables S1 and S2). Excessive mutations per step would erase any evolutionary information embedded in the sequence, whereas too few mutations limit the overall design efficiency. To restrict the mutational load, a weight was added to RosettaDesign that penalizes mutations relative to the input structure. For ProteinMPNN, this was achieved similarly by adding a bias for the input sequence. Consequently, mutation rate, Boltzmann selection temperature, and design strategy are key tunable parameters for optimizing AI.zymes’ performance.

Evolutionary Design Outperforms Non-evolutionary Design

The promiscuous Kemp eliminase activity of ketosteroid isomerase (KSI) was selected as a target to benchmark AI.zymes.^[28] The Kemp elimination, the base-catalyzed C–H deprotonation of benzisoxazoles, is an established benchmark for de novo enzymes (Figure 2a).^[3,29–31] KSI is a dimeric enzyme that promiscuously catalyzes the Kemp elimination of 5-nitrobenzisoxazole. The Kemp eliminase activity of *Comamonas testosteroni* KSI has previously been improved by introducing a D38N mutation, removing one of the two active-site aspartates that could act as a catalytic base.^[28] Our initial characterization confirmed the Kemp eliminase activity of KSI D38N, referred to as WT hereafter, although size-exclusion chromatography revealed that the protein formed an oligomeric mixture (Figure 4c). Therefore, AI.zymes was tasked with both enhancing activity and creating a thermostable and monomeric enzyme.

To benchmark AI.zymes, initial design runs were performed solely using RosettaDesign to optimize the active-site residues and ESMFold and RosettaRelax for structure prediction. After fine-tuning the Boltzmann selection temperature and mutation rate (Table S1), computational evolution outperformed non-evolutionary design in terms of interface and stability score, while exploring substantially broader sequence

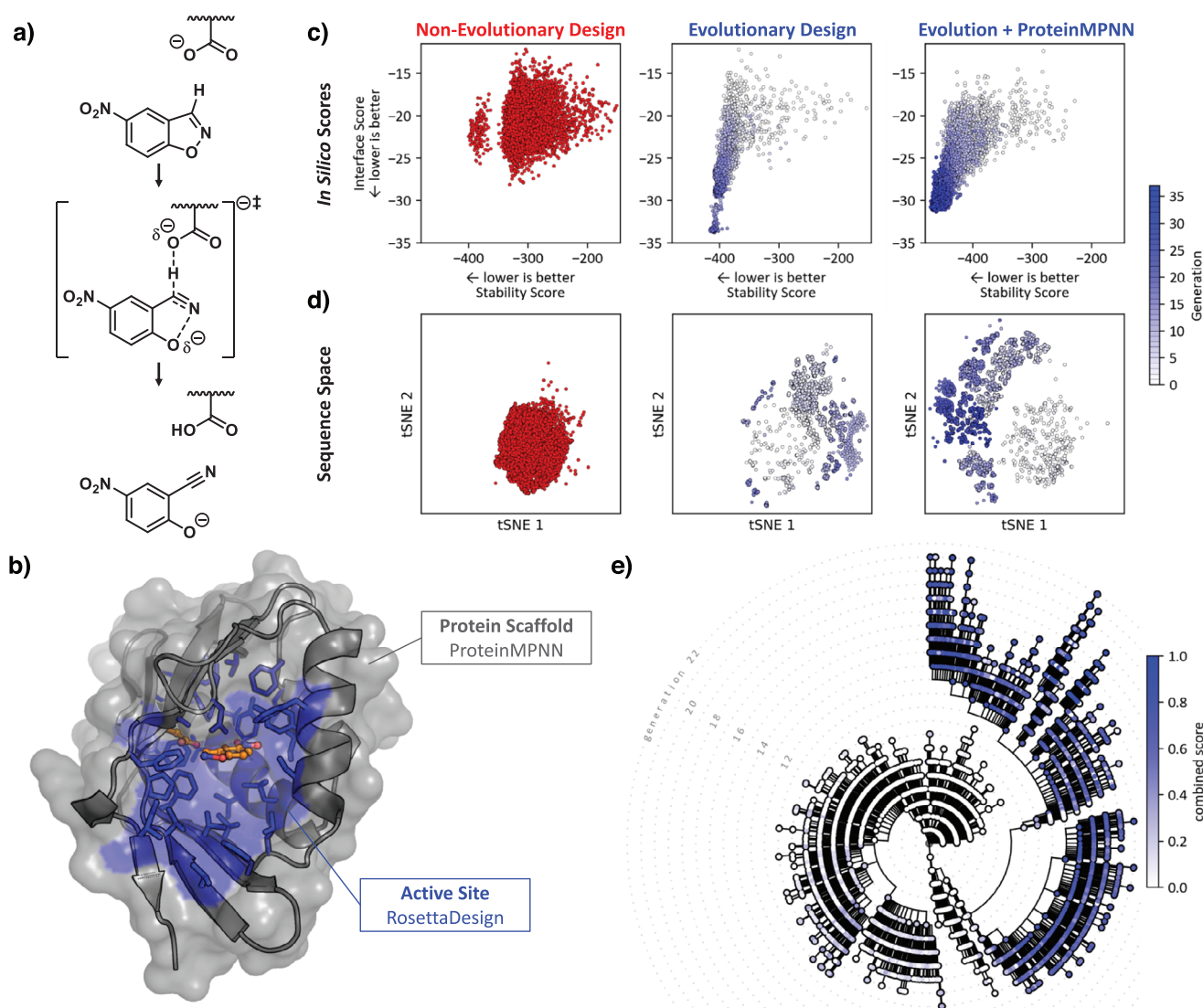


Figure 2. Evolutionary design outperforms non-evolutionary design. a) AI.zymes was benchmarked by improving the Kemp eliminase activity of KSI. b) The active site of KSI was targeted by RosettaDesign (blue) and the protein scaffold was redesigned with ProteinMPNN (grey; chemical transition state and catalytic base Asp99: orange; structure based on PDB ID 1OHP).^[27] Compared to non-evolutionary design (red), evolutionary design (blue) achieved c) higher in silico scores and d) explored a broader sequence space (tSNE analysis of ESM2-embedded sequences). e) Phylogenetic tree tracing ancestors during computational evolution. The combined score represents the standardized mean of all scores targeted during design.

space (Figure 2c+d). Integration of ProteinMPNN, which excels in creating thermostable proteins,^[9] into AI.zymes further improved the designs by increasing the stability score and expanding the explored sequence space. AI.zymes readily tracks the evolutionary design history, which enabled the construction of a phylogenetic tree that maps ancestral variants to their descendants. This phylogenetic tree clearly visualizes the incremental improvements achieved during design (Figure 2e).

Evolutionary Design of Electric Fields

Catalytic electric fields are critical in stabilizing transition states and promoting enzyme activity (Figure 3a).^[20,21] Despite a few notable successes,^[22,32] the computational

design of catalytic electric fields has been challenging. The limited k_{cat} values of designer enzymes^[3,15,16] can probably be attributed partly to the neglect of electrostatic catalysis. To address this gap in current design methodologies, we integrated FieldTools, which we recently developed to assess electric fields from MD trajectories,^[19] into AI.zymes (Figure S1).

Electric field design was aimed at improving the effective field along the scissile C–H bond. Importantly, neither RosettaDesign nor ProteinMPNN can explicitly design catalytic electric fields. AI.zymes achieves field optimization by combining protein design algorithms unaware of electric fields with Boltzmann selection for variants with stronger catalytic fields (Figure 3b). The most effective fields emerged from designs combining Rosetta and ProteinMPNN, rather than Rosetta alone, highlighting the role of remote scaffold mutations. These results suggest that electric field optimization—a

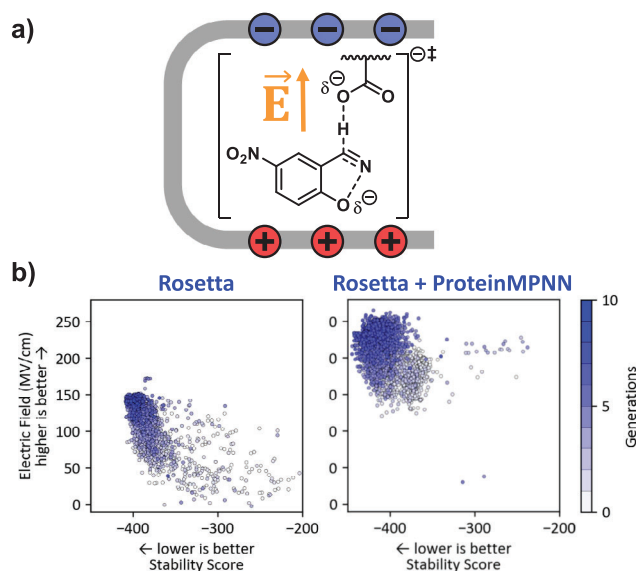


Figure 3. Designing catalytic electric fields. a) Electric fields (E , orange) quantify electrostatic catalysis. The electric field reflects the effective field along the scissile C—H bond. b) Design runs with ProteinMPNN and RosettaDesign result in higher catalytic electric fields than those using only RosettaDesign.

significant advancement in enzyme design—can be achieved with design methods not tailored for field effects.

Molecular Dynamics Screening

We previously employed short (10 ns) molecular dynamics (MD) simulations to evaluate the conformational stability of enzyme-bound ligands.^[23] Importantly, these short simulations are not intended to identify the best designs but rather to pinpoint bad variants that rapidly adopt nonproductive conformations. Here, the top 96 designs underwent MD screening (Figure S2).^[33–41] Variants were filtered based on 1) ligand flexibility, 2) ligand-base distance, 3) electric field strength, 4) preorganization of the apo state, and 5) active-site hydration. MD filtering identified 23 variants, from which 7 were chosen for experimental testing after visual inspection of the active-site geometry.

Experimental Validation

Experimental validation is crucial to the design of functional proteins. Activity assays at the highest possible substrate concentration (1.5 mM) revealed that 1 out of 7 tested variants, K3, surpassed WT activity, whereas the others performed similarly or worse (Table S3).^[42] pH-Rate profiles revealed that K3 exhibited a 7.7-fold increase in maximum $k_{\text{cat}}/K_{\text{M}}$ compared to WT, along with a shift in the general base pK_{a} from 8.1 to 8.4 (Table 1 and Figure 4a,b). Activity improvements were likely driven by increases in k_{cat} as full Michaelis–Menten kinetics at pH 7.0 showed that the rate enhancements were almost entirely due to higher turnover rates (Figure S3c). Although evolutionary design altered 45%–55% of the protein sequence, only 7 variants had to be tested to find

Table 1: Kinetics of WT and K3.

Variant	$(k_{\text{cat}}/K_{\text{M}})_{\text{max}}$ ($\text{M}^{-1} \text{s}^{-1}$) ^a	pK_{a} ^a
WT	1200 ± 200	8.1 ± 0.1
K3	9200 ± 1800	8.4 ± 0.1

^a) Maximum $k_{\text{cat}}/K_{\text{M}}$ determined from pH-rate profiles. Error bars reflect the standard error of four replicates.

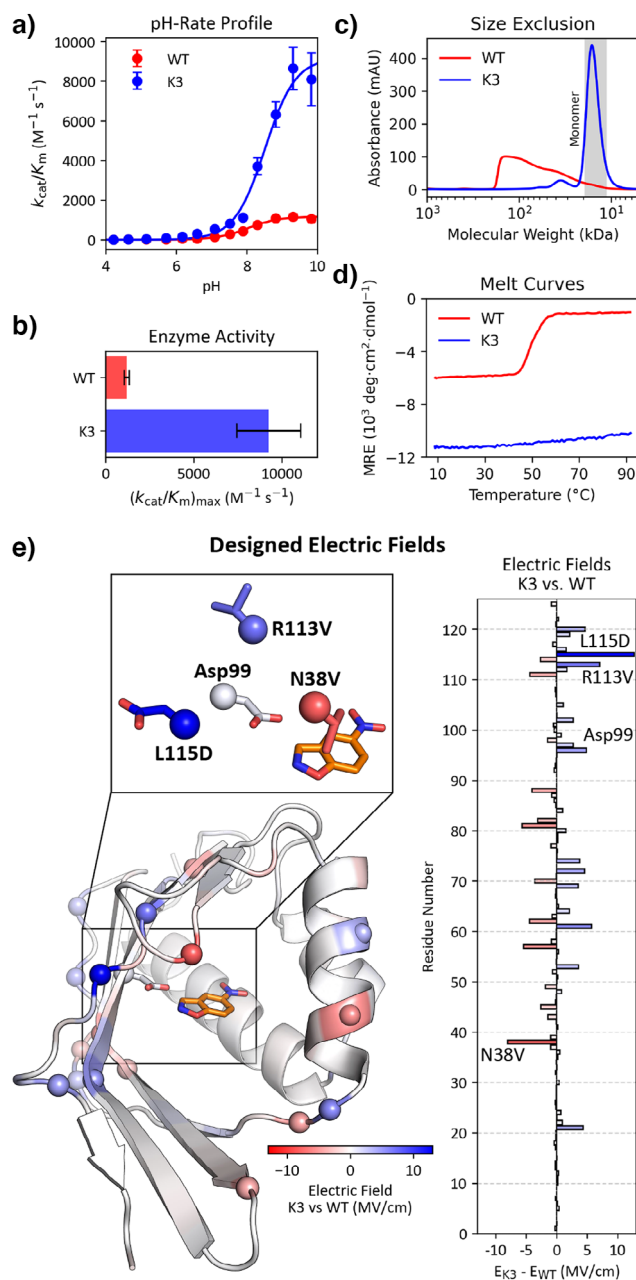


Figure 4. Evolutionary design improved enzyme activity. a) and b) pH-Rate profiles reveal that K3 is up to 7.7 times more active than WT. Error bars reflect the standard error of four replicates. c) K3 is monomeric, whereas WT forms an oligomeric mixture. d) Although WT unfolds at 49.8 °C, K3 is stable up to 95 °C. e) Per-residue comparison of the electric field in K3 versus WT reveals the origins of the increased field (increased field: blue; decreased field: red).

one with improved activity. This contrasts with computational or machine-learning-assisted enzyme engineering campaigns that typically involve extensive experimental screening.

From a structural perspective, design aimed at creating a stable monomeric protein. Replacing hydrophobic residues with hydrophilic ones at the KSI dimer interface afforded a predominantly monomeric variant, as indicated by size-exclusion chromatography (Figures 4c, S3b, and S4). This improved structural definition was corroborated by AlphaFold3 predicting K3 to be a stable monomeric state, whereas WT could adopt multiple oligomeric forms (Figure S5). Furthermore, circular dichroism (CD) melting curves support that K3 is highly thermostable and does not unfold up to 95 °C, whereas WT had a melting temperature of 49.8 °C. Notably, K3 exhibited stronger CD signals, further supporting that multiobjective optimization enhanced activity while affording a well-behaved and thermostable catalyst (Figure S3d).

Electric Field Optimization

Inspection of the mutations between WT and K3 revealed that the improvements in the total electric field arose from numerous small contributions (Figure 4e). For instance, L115D introduced a negative charge that aids deprotonation, whereas R113V removes an anticatalytic positive charge in the WT. Interestingly, AI.zymes also introduced mutations such as N38V that negatively impact the electric field but enhance substrate binding by increasing the binding-site hydrophobicity. Notably, AI.zymes optimized the electric fields without relying on design tools tailored to design electrostatically beneficial mutations. This suggests that AI.zymes can optimize any aspect of enzyme function—even those that cannot be directly targeted by design—provided the desired trait can be scored and used in Boltzmann selection.

To probe how protein dynamics affect electrostatics, we analyzed K3 by 10 ns MD simulations. These analyses revealed an increase in the electric field of 19 MV cm⁻¹ compared to WT. The improvement was slightly lower than that calculated from the static structures (23 MV cm⁻¹), underscoring the importance of dynamics. In fact, repeated scoring of WT and K3 through ESMfold and RosettaRelax reproduced the score fluctuations seen in MD (Figure S6). To assess conformational stability over longer timescales, we performed eight replicate 1 μs simulations of WT and K3. As expected, the ligand remained stably bound in all K3 trajectories, whereas in several WT replicates, the ligand adopted alternative conformations or dissociated from the active site (Figure S7). While these findings validate the results obtained from static structures, they also suggest that including the effects of dynamics on all calculated scores could further improve the design outcome.

Conclusion

Here, we developed AI.zymes, a modular platform that advances enzyme design through evolutionary multiobjective optimization. AI.zymes was successfully utilized to enhance

the promiscuous Kemp eliminase activity of KSI by optimizing transition-state recognition, catalytic geometry, and electric fields while also boosting protein stability. Integrating a range of cutting-edge computational tools, including RosettaDesign, ProteinMPNN, ESMFold, and FieldTools,^[19] enabled the efficient exploration of the enzyme fitness landscape, resulting in a 7.7-fold improved activity after experimentally testing just 7 variants.

A key advantage of evolutionary design is its ability to target mutagenesis toward functional sequence space, which increases design efficiency compared to non-evolutionary methods. Increased efficiency is valuable for both large-scale design projects generating millions of designs and small-scale efforts constrained by limited computational resources. Boltzmann selection further enhances efficiency by operating on the entire design pool rather than using rounds-based approaches like experimental directed evolution. This approach maximizes resource utilization by continuously running a fixed number of designs without waiting for one round to finish before starting the next. Additionally, AI.zymes capability for multiobjective optimization—even for traits not directly addressed by the design algorithms, such as electrostatic catalysis—further improves functionality without increasing computational costs.

Outlook

Creating enzymes remains one of the major challenges in computational protein design. Although recent design campaigns succeeded in generating some activity by building protein scaffolds around active-site models,^[15,16] designing highly efficient enzymes will probably require extending beyond the design of defined protein-transition state complexes. Protein dynamics and electrostatic catalysis should be included during design—not as filters after the design—to more efficiently account for catalytically relevant conformational sampling and long-range effects within the protein scaffold.^[3,17,18,20,22,32]

The modularity of AI.zymes will facilitate the integration of cutting-edge bioengineering tools. For instance, algorithms such as LigandMPNN, RFdiffusionAA, and AlphaFold3, which can account for nonprotein ligands, could be integrated to model ligand-bound structures with increased accuracy.^[14,43,44] In addition, methods such as MD simulations or deep-learning tools like ChemNet^[45] could further enhance design by probing conformational dynamics. Our MD screening narrowed the range of design hits from 96 to 7 variants, suggesting that short simulations (≤10 ns) can eliminate designs with nonproductive conformations. In total, 10⁴ designs were created during the design of K3, a number for which short MD simulations would be computationally feasible.

AI.zymes successfully integrates catalytic electric fields into design. While AI.zymes optimized the field strength along the scissile C–H bond, it did not account for charge delocalization within the substrate—a feature particularly relevant for Kemp eliminases.^[46] To address this and better capture electrostatic effects across the whole substrate, AI.zymes

may be extended to account for electric field reorganization along all bonds and across the full reaction coordinate. A more refined description of electrostatic catalysis could support the design of more efficient artificial enzymes.

In addition to relying on physics- or AI-based design tools, AIzymes could be integrated with data-driven reinforced machine learning models to help guide exploration toward more active regions of the fitness landscape. Similar strategies have been applied in machine-learning-assisted directed evolution for experimental enzyme optimization^[47] and in training deep-learning models on sequence databases to develop new biocatalysts.^[48–51]

Finally, AIzymes could also be interfaced with de novo enzyme design strategies to generate biocatalytic activity entirely from scratch. In such workflows, the initial design step could follow the theozyme approach, either by docking active-site models into existing protein scaffolds^[52,53] or by constructing de novo protein backbones around theozymes.^[15,16] The resulting designs could then be further optimized through in silico evolution using AIzymes.

Simultaneously optimizing stability, preorganization, electrostatics, and dynamics is challenging for any single design algorithm. AIzymes addresses this with a modular, evolutionary framework integrating diverse computational tools. Moreover, evolutionary design allows properties beyond those explicitly targeted by the employed design algorithms to be improved. Its flexibility positions AIzymes as a general platform for computational protein evolution, with the potential to accelerate biocatalyst development for industry, medicine, and synthetic biology.

Author Contributions

H.A.B., L.M., J.N., and A.L. developed the code with support from T.S.C. and V.D. L.M. performed the computational design of K3. A.L. performed the MD simulations. L.M., V.D., and M.M.W. recorded the experimental data. H.A.B. and L.M. wrote the manuscript with input from all coauthors.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

AIzymes is available at <https://github.com/bunzela/AIzymes>. All data and methods required to reproduce this work are available at the University of Bristol data repository (<https://data.bris.ac.uk/data/dataset/fvbvxg89ldwn2kv5ecxba9b7m>) and Supporting Information.

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